

RHYL, United Kingdom

WMO: 033130

Lat: 53.2592N

Lon: 3.5089W

Elev: 76

StdP: 100.42

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.4	(c) -1.0	(d) -5.7	(e) 2.4	(f) -0.1	(g) -4.4	(h) 2.6	(i) 0.7	(j) 12.6	(k) 8.6	(l) 11.5	(m) 8.9	(n) 2.3	(o) 210	(p) 0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	23.6	17.6	21.7	16.9	20.2	16.2	18.5	22.2	17.6	20.7	16.7	19.5	3.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	12.3	20.2	16.2	11.6	19.2	15.5	11.1	18.4	52.6	22.2	49.7	20.8	47.2	19.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	8.9	7.8	-4.5	26.9	1.7	2.0	-5.7	28.3	-6.8	29.5	-7.7	30.6	-9.0	32.1
			-5.1	21.0	1.9	2.1	-6.4	22.5	-7.5	23.7	-8.6	24.9	-9.9	26.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	5.4	5.6	6.8	8.7	11.3	14.0	15.8	15.7	13.8	11.2	8.1	6.0
	DBStd	4.63	3.07	2.96	2.83	2.74	2.65	2.27	2.20	2.04	2.01	2.62	2.79	3.38
	HDD10.0	660	142	125	104	57	15	1	0	0	1	17	69	129
	HDD18.3	2969	399	357	358	291	218	131	84	86	136	220	307	383
	CDD10.0	745	1	2	5	17	55	122	181	177	115	55	12	4
	CDD18.3	13	0	0	0	0	0	2	6	4	1	0	0	0
	CDH23.3	74	0	0	0	0	3	13	34	22	2	0	0	0
	CDH26.7	9	0	0	0	0	0	1	6	3	0	0	0	0
Wind	WSAvg	3.5	4.3	4.2	4.1	3.5	3.1	3.1	3.0	3.0	3.0	3.4	3.6	4.0
Precipitation	PrecAvg	804	75	61	46	48	55	56	62	68	68	86	88	90
	PrecMax	1099	135	138	103	98	108	138	127	142	143	162	217	199
	PrecMin	577	0	16	8	2	18	13	16	13	4	21	33	28
	PrecStd	124	38	32	25	26	24	33	28	29	34	38	42	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.1	15.9	20.4	23.0	25.2	26.9	26.3	22.6	19.1	15.5	14.0
		MCWB	11.0	10.7	11.0	13.8	17.0	18.2	18.9	18.8	17.5	15.5	13.4	11.9
	2%	DB	11.8	12.0	13.4	16.3	19.8	22.0	23.6	22.4	20.3	17.3	14.3	12.8
		MCWB	10.0	9.9	10.2	11.8	15.0	16.8	17.7	17.7	16.1	14.6	12.6	11.3
	5%	DB	10.9	11.0	12.1	14.3	17.7	19.8	21.7	20.5	18.7	16.3	13.3	11.9
		MCWB	9.2	9.0	9.6	10.7	13.7	15.5	16.9	16.8	15.3	13.8	11.5	10.2
	10%	DB	10.0	10.0	11.0	12.9	15.9	18.2	20.0	19.2	17.4	15.3	12.3	10.8
		MCWB	8.3	8.1	8.7	10.0	12.5	14.6	16.2	16.0	14.5	13.0	10.5	9.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.2	12.1	14.0	18.0	19.3	20.0	19.7	18.4	16.2	13.9	12.3
		MCDB	12.8	13.0	14.6	19.1	22.7	23.1	24.3	23.9	21.5	18.1	15.2	13.6
	2%	WB	10.3	10.2	10.7	12.4	15.6	17.2	18.5	18.3	16.9	15.2	12.9	11.4
		MCDB	11.6	11.8	12.7	15.6	19.1	20.9	22.5	21.5	19.4	17.1	14.2	12.7
	5%	WB	9.4	9.3	9.7	11.3	14.2	16.0	17.5	17.4	15.9	14.2	11.8	10.5
		MCDB	10.7	10.9	11.8	13.8	16.9	19.1	20.8	20.0	18.0	16.0	13.2	11.8
	10%	WB	8.5	8.3	9.0	10.2	13.0	15.1	16.5	16.6	14.9	13.2	10.7	9.3
		MCDB	9.9	9.9	10.8	12.5	15.4	17.9	19.5	18.8	17.0	15.0	12.1	10.7
Mean Daily Temperature Range	5% DB	MDBR	4.6	4.9	5.8	6.6	7.2	7.1	6.9	6.5	6.4	5.7	4.8	4.7
		MCDBR	5.3	5.8	7.1	8.7	10.1	10.0	9.9	8.5	8.4	6.5	5.6	5.3
		MCWBR	5.1	4.9	5.1	5.6	6.4	6.2	5.6	5.1	5.4	4.7	4.9	4.9
	5% WB	MCDBR	5.3	5.6	6.5	8.0	9.3	8.9	8.7	7.5	7.2	6.0	5.4	5.4
		MCWBR	5.3	5.0	5.0	5.5	6.3	6.0	5.6	4.9	5.3	4.9	5.0	5.1
Clear-Sky Solar Irradiance	taub	0.330	0.342	0.373	0.395	0.400	0.393	0.394	0.378	0.368	0.351	0.327	0.319	
	taud	2.299	2.356	2.288	2.253	2.291	2.340	2.345	2.401	2.424	2.423	2.363	2.273	
	Ebn at Noon	630	755	804	833	849	858	850	844	806	736	633	561	
	Edh at Noon	62	79	103	121	123	118	116	103	90	73	58	53	
All-Sky Solar Radiation	RadAvg	0.65	1.32	2.48	3.83	4.85	5.10	4.81	3.95	2.85	1.58	0.79	0.49	
	RadStd	0.06	0.14	0.22	0.36	0.34	0.48	0.48	0.26	0.21	0.14	0.08	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page